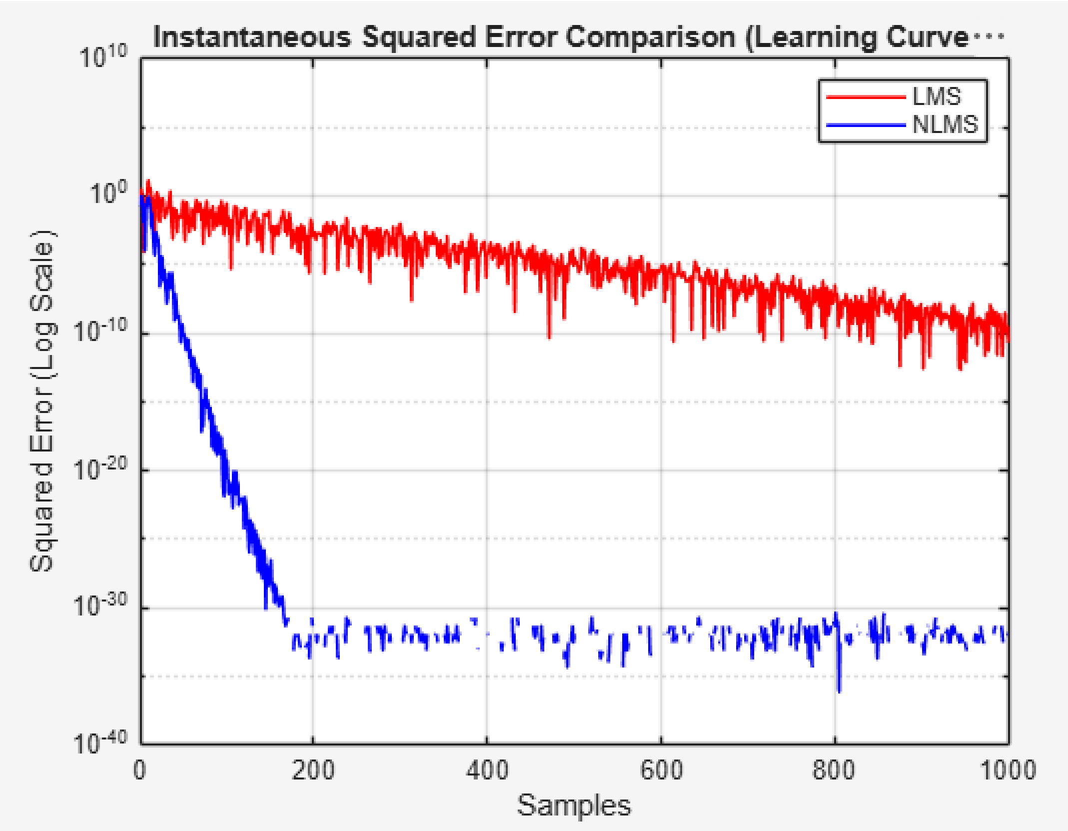
9. Implementation and Performance Analysis of LMS and NLMS Algorithms for Adaptive Echo Cancellation

/MATLAB Drive/Dsp_9.m

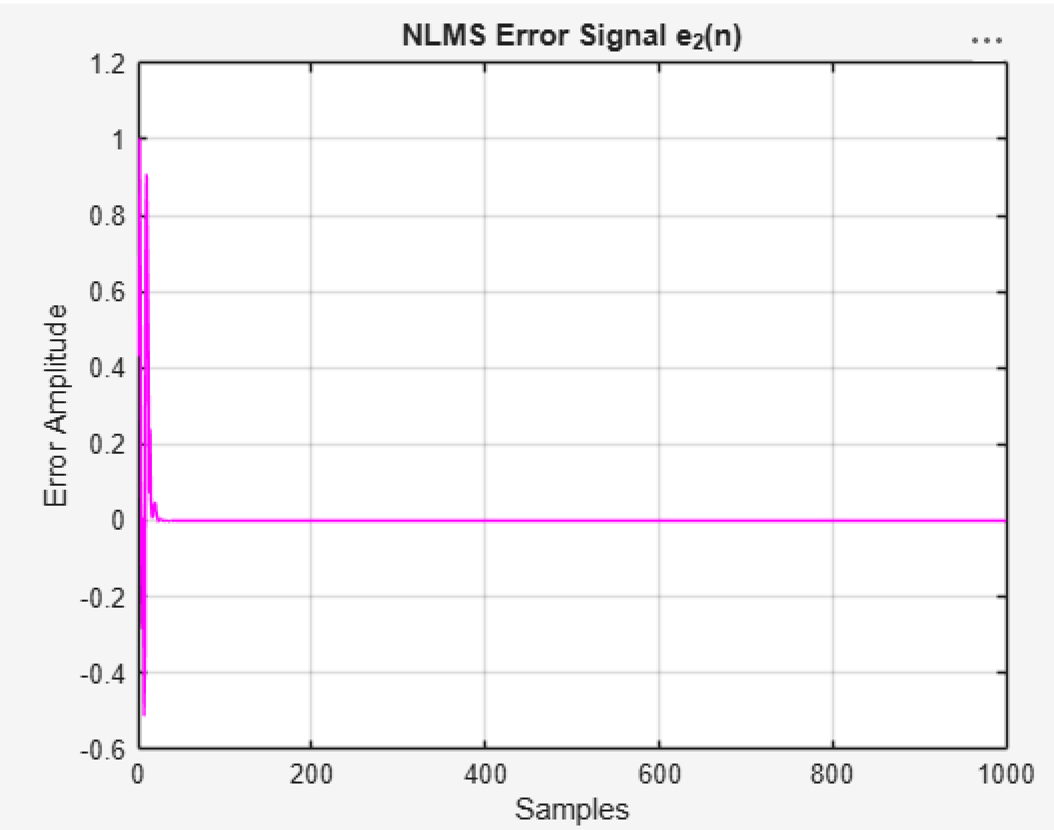
```
clc; clear; close all;

%% Signal Generation & System Setup
N = 1000;                       % Number of samples
x = randn(1, N);                % Input signal (white Gaussian noise)

% Echo Path / Unknown System Impulse Response: H(z) = 0.8 + 0.5 z^-1 + 0.3 z^-2
h_true = [0.8, 0.5, 0.3];
d = filter(h_true, 1, x);       % Desired signal (echoed output)

%% Adaptive Filter Initialization
filterOrder = 3;                % Length of adaptive filter
w_lms = zeros(filterOrder, 1);  % Weights for LMS
w_nlms = zeros(filterOrder, 1); % Weights for NLMS

mu_lms = 0.01;                  % LMS Step Size
mu_nlms = 0.5;                  % NLMS Step Size
eps_nlms = 1e-4;                % Regularization factor to avoid division by zero

e1 = zeros(1, N);               % Error signal for LMS
e2 = zeros(1, N);               % Error signal for NLMS

x_buf = zeros(filterOrder, 1);  % Delay buffer for input vector

%% LMS and NLMS Adaptive Filtering Loop
for k = 1:N
    % Update buffer with current input sample
    x_buf = [x(k); x_buf(1:end-1)];

    % --- LMS Algorithm ---
    y1 = w_lms' * x_buf;                % Filter output
    e1(k) = d(k) - y1;                  % Error
    w_lms = w_lms + mu_lms * e1(k) * x_buf; % Weight update

    % --- NLMS Algorithm ---
    y2 = w_nlms' * x_buf;               % Filter output
    e2(k) = d(k) - y2;                  % Error
    norm_factor = (x_buf' * x_buf) + eps_nlms; % Normalized energy
    w_nlms = w_nlms + (mu_nlms / norm_factor) * e2(k) * x_buf; % Weight update
end
```

```matlab
%% Plotting Results

% 1. Input and Echo Signal
figure('Name', 'Input vs Echo', 'NumberTitle', 'off');
plot(x, 'b');
hold on;
plot(d, 'r');
grid on;
legend('Input', 'Echo');
title('Input and Echo Signal');
xlabel('Samples');
ylabel('Amplitude');

% 2. LMS Error Output
figure('Name', 'LMS Error Output', 'NumberTitle', 'off');
plot(e1, 'g');
grid on;
title('LMS Error Signal e_1(n)');
xlabel('Samples');
ylabel('Error Amplitude');

% 3. NLMS Error Output
figure('Name', 'NLMS Error Output', 'NumberTitle', 'off');
plot(e2, 'm');
grid on;
title('NLMS Error Signal e_2(n)');
xlabel('Samples');
ylabel('Error Amplitude');

% 4. Squared Error (Instantaneous MSE) Comparison
figure('Name', 'MSE Comparison', 'NumberTitle', 'off');
semilogy(e1.^2, 'r', 'LineWidth', 1.2);
hold on;
semilogy(e2.^2, 'b', 'LineWidth', 1.2);
grid on;
legend('LMS', 'NLMS');
title('Instantaneous Squared Error Comparison (Learning Curves)');
xlabel('Samples');
ylabel('Squared Error (Log Scale)');
```

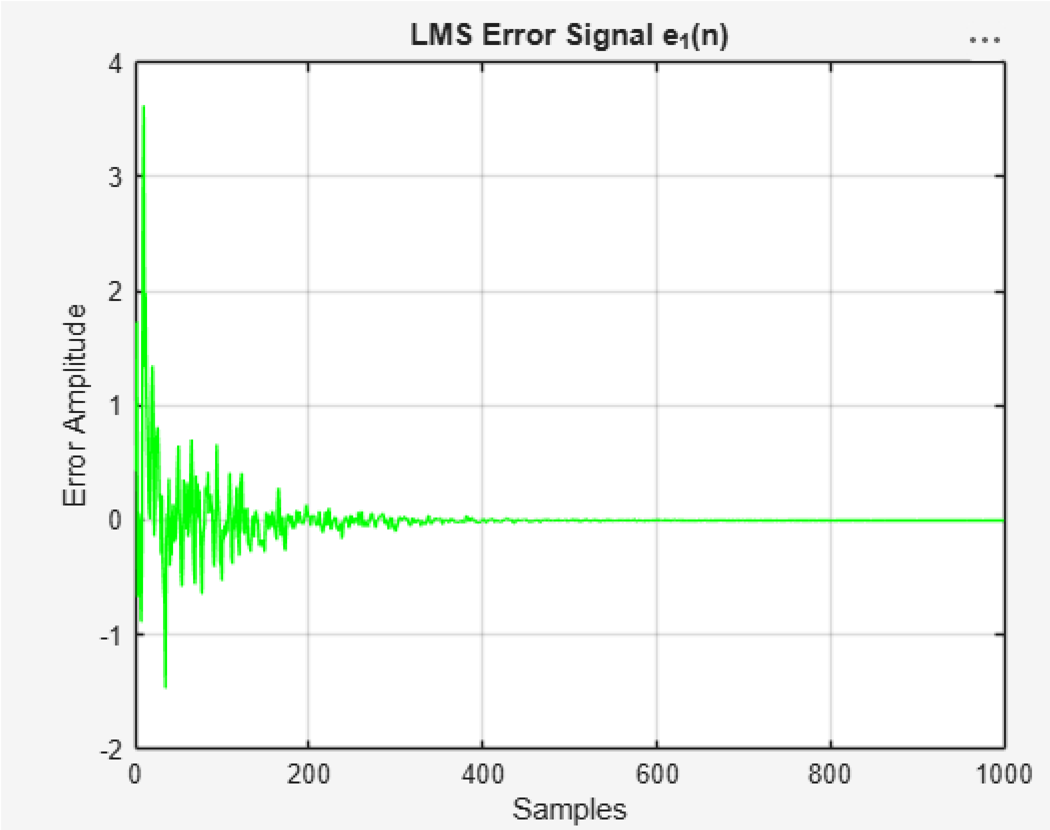
Output:

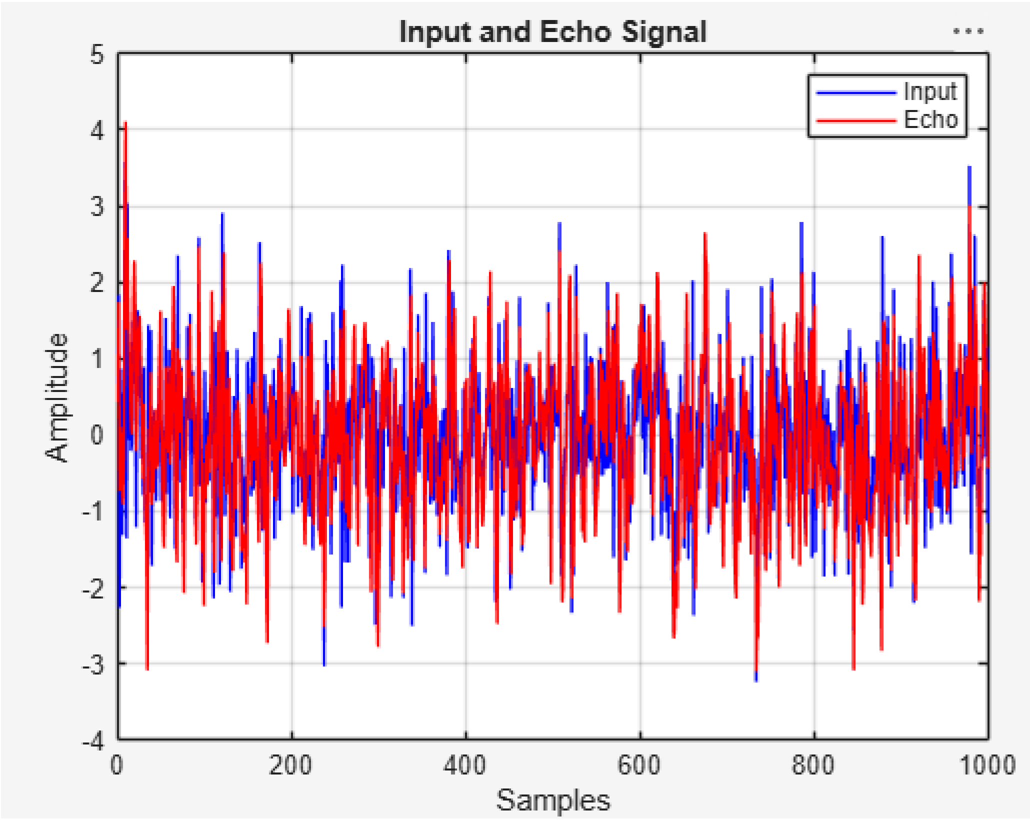


Instantaneous Squared Error Comparison (Learning Curve
LMS
NLMS
Squared Error (Log Scale)
10^10
10^0
10^-10
10^-20
10^-30
10^-40
0
200
400
600
800
1000
Samples



NLMS Error Signal e2(n)
Error Amplitude
1.2
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
0
200
400
600
800
1000
Samples


LMS Error Signal e1(n)
Error Amplitude
Samples
4
3
2
1
0
-1
-2
0
200
400
600
800
1000


Input and Echo Signal
Input
Echo
Amplitude
Samples
5
4
3
2
1
0
-1
-2
-3
-4
0
200
400
600
800
1000